

RAJA ANTON DIGNAS

RESEARCH ASSISTANT | DATA ANALYSIS & NLP

Mocuba, Zambezia, Mozambique | +258 84 911 55 98 - 86 487 48 75 |

rajadignas@gmail.com | <http://linkedin.com/in/raja-dignas-117759208>

PROFESSIONAL PROFILE

Master's degree in Artificial Intelligence specializing in Natural Language Processing and Bachelor's degree in Information Systems and Software Engineering.

Over 5 years of academic, associative, and research experience in an international and multicultural environment in Algeria.

Experience in data analysis, ETL, data visualization, predictive modeling, and dashboard development to support decision-making.

Motivated to apply my experience in data analysis and predictive modeling to the challenges of agriculture and climate in Sub-Saharan Africa.

Strong experience in leadership, project coordination, and institutional communication with diverse audiences and official institutions.

Proactive, autonomous, and rigorous in handling sensitive data.

PROFESSIONAL & ACADEMIC EXPERIENCE

Research Assistant & Data Analyst | Master's Project | 2024-2025

- 1. Large-Scale Text Data Analysis:** Developed a Sentiment Analysis model using CamemBERT on +10,000 customer comments. Achieved 90% Accuracy and 87% F1-score.
- 2. Data Collection & Processing:** Web scraping, data cleaning, and harmonization of multilingual data PT/EN/AR → FR. Automation that reduced manual analysis time by 87%.
- 3. KPIs & Visualization:** Defined performance indicators and created dashboards with Matplotlib to identify trends and critical points in time series.
- 4. Predictive Modeling:** Identified key risk factors from complex datasets to support decision-making.

ETL Technician & Data Quality | R&D Project | 2024-2025

- 1. Data Pipeline:** Built ETL pipelines with Python, Pandas, and SQL for the organization and quality control of large volumes of data.
- 2. Validation & Documentation:** Model validation using F-score, Precision@k metrics. Code versioning and documentation with Git.
- 3. Database Management:** CRUD operations in MySQL and MongoDB with a focus on data integrity, security, and traceability.

Radio Columnist | Ifriky FM | Algeria | Mar - Jul 2025

- 1. Institutional Communication:** Weekly production and presentation of columns. Synthesis and popularization of technical and scientific topics for the general public.
- 2. Multicultural Interaction:** Media coverage and interviews with official representatives of embassies and institutions at diplomatic events in Algiers.

ASSOCIATIVE COMMITMENT & LEADERSHIP

National Coordinator of an Organization | Algeria | 2022-2025

Team Management: Directed a national organization with a regional council and department heads. Training and mentoring of leaders.

Project Management* Organized quarterly seminars with +200 people and an annual event with +500 people. Managed logistics and budget.

Coordination: Planning, internal communication, and representation of the organization to partners.

President | Central African Students' Association in Algeria (ULECA)| 2021-2022

Institutional Relations: Liaison between a diplomatic representation and the student community. Management of sensitive processes: enrollments, transfers, administrative support.

Leadership: Managed the executive secretariat and represented the interests of +300 students to university authorities.

Deputy Secretary General | Community of African Foreign Students of Blida | 2020-2021

Reception & Integration: Coordinated the reception of +500 African scholarship students. Support with university housing and faculty registrations, administrative processes.

Associative Life: Organized cultural activities to promote integration in a multicultural Francophone, Lusophone, Anglophone, and Arabophone environment.

KEY RESEARCH-ORIENTED PROJECTS

1.Customer Satisfaction Data Analysis | 2024

Applied NLP techniques and predictive modeling to analyze +10k unstructured data points and identify risk factors. Skills transferable to the analysis of farmer feedback.

2.Semantic Search Engine for Scientific Documents | 2024

Designed a search engine using LSI. Processed queries and calculated similarity between documents. Evaluation: F-score, Precision@K, DCG. Experience in managing research corpora.

3.Relational Database Management System | 2024

Modeling in SQL Server. Implementation of business rules, integrity constraints, and data quality validation tests. Applicable to agricultural databases.

4. Document Classification & Matching | 2023

NLP preprocessing with NLTK, TF-IDF vectorization, and KNN classification. Model evaluation with precision and recall.

EDUCATION

Master's in Artificial Intelligence - NLP Specialization | University of Blida 1, Algeria | 2025

Thesis: "*Sentiment Analysis on the Service Quality of Mobile Operators in Algeria*"

Bachelor's in Information Systems and Software Engineering | University of Blida 1, Algeria | 2023

International Experience: Over 5 years in university residence with scholarship students from more than 20 African countries.

TECHNICAL SKILLS

Data Analysis & BI: Python, Pandas, NumPy, Advanced SQL, ETL, KPI, Exploratory Data Analysis, Matplotlib, Seaborn, Advanced Excel

AI/ML & NLP: Machine Learning, Predictive Modeling, Sentiment Analysis, BERT, CamemBERT, Text Processing

Databases: Oracle, SQL Server, MySQL, MongoDB

Tools: Git, Technical Documentation, Report Writing

Application Areas: Agricultural data analysis, Climate time series analysis, Research database management

SOFT SKILLS

Leadership | Project Management | Institutional Communication | Multicultural Teamwork | Negotiation | Technical Communication | Scientific Rigor

LANGUAGES^m

French: Native | **Portuguese:** Intermediate | **English:** Fluent | **Arabic:** Basic

Availability: Immediate for Maputo

PROFESSIONAL REFERENCES

1.Dr Mezzi Melyara

Head of NLP Specialization | Lecturer and Researcher | University of Blida 1, Algeria | Email: mezzi.melyara@gmail.com

2.Dr TEBBI Hanane

Lecturer and Researcher | Department of Computer Science | University of Blida 1, Algeria | Email: tebbi_hanane@yahoo.com

3.Dr Kameche Abdallah

Lecturer and Researcher, Deputy Head of Computer Science Department | University of Blida 1, Algeria | Email: kameche.abdallah@gmail.com